

CYCLISTIC BIKE-SHARE

Converting Casual Riders: A Data-Driven Campaign Strategy

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Executive Summary

Cyclistic is a Chicago-based bike-share company with two distinct rider segments: annual members and casual riders. While members generate predictable, year-round revenue, casual riders pay per trip and represent the highest opportunity for conversion.

This analysis examined 5,691,593 rides recorded between May 2025 and April 2026 — 3,658,607 member rides and 2,032,986 casual rides. Using R for data processing and Tableau for visualization, four behavioral patterns were identified that reveal not only how the two groups differ, but specifically when and where Cyclistic should invest in conversion campaigns.

The core finding: casual riders are leisure users. They ride on weekends, in the afternoon, take longer trips (averaging 19 minutes vs. 12 minutes for members), and cluster near tourist areas. This behavioral profile creates a clear and actionable conversion opportunity during spring, when casual demand accelerates before its summer peak.

Business Problem

Cyclistic's finance team has determined that annual members are significantly more profitable than casual riders. The marketing director, Lily Moreno, has set a clear objective:

"Design marketing strategies aimed at converting casual riders into annual members."

To achieve this, the marketing team needs to understand how annual members and casual riders use Cyclistic bikes differently — and use those insights to target the right riders, at the right time, in the right place.

Data & Methodology

Dataset

- Source: Cyclistic historical trip data (publicly available dataset provided by Motivate International Inc. under license)
- Period analyzed: May 2025 to April 2026 (12 months)
- Total rides after cleaning: 5,691,593 (3,658,607 member | 2,032,986 casual)
- Fields used: start/end timestamps, station names, GPS coordinates, ride type, rider type (casual vs. member)

Data Cleaning (R / tidyverse)

- Removed rows with null values in started_at, ended_at, member_casual, and rideable_type
- Calculated ride duration in minutes using difftime()
- Removed invalid rides: duration ≤ 0 minutes or $> 1,440$ minutes (24 hours)
- Engineered features: day of week, month, hour of day, season

Tools

- R (tidyverse, lubridate) — data processing and CSV exports
- Tableau Public 2026.1.1 — visualization and dashboard
- 12 CSV files processed and combined into a single analytical dataset

Data Limitations

The following limitations apply to this analysis and should be considered when interpreting the results:

- Pricing data not included: Cyclistic did not provide pricing information in the dataset. As a result, no exact break-even analysis (e.g., how many casual rides equal the cost of an annual membership) was possible. Recommendations referencing financial value are based on structural pricing logic — not specific fare calculations.
- No demographic data: To protect rider privacy, all personally identifiable information was removed from the source data. Age, gender, and residential information are not available, limiting demographic segmentation.
- Station data gaps: A portion of rides had missing station names or GPS coordinates. These rows were retained for temporal and behavioral analysis but excluded from geographic visualizations.
- Data is historical: This dataset covers May 2025 to April 2026. Behavioral patterns may shift over time due to infrastructure changes, pricing adjustments, or external factors not reflected in this data.

Key Findings

Finding 1: Behavior Reveals Intent

The behavioral fingerprint — a heatmap of rides by hour of day and day of week — reveals a stark contrast between the two groups.

Members: Commuters

Peak activity at 8am and 5–6pm, Monday through Friday.

Low activity on weekends and outside of rush hours.

Pattern is consistent with using Cyclistic as a daily transportation tool.

Casual Riders: Leisure Users

Peak activity in the afternoon (12pm–6pm) across all days.

Significantly higher activity on Saturday and Sunday.

Pattern is consistent with recreational use — tourism, exercise, social outings.

This distinction is the analytical foundation for all three recommendations below. Casual riders are not potential commuters — they are leisure users who need a value proposition built around flexibility and seasonal value.

Finding 2: Casual Riders Take Longer, More Expensive Trips

Duration analysis quantifies the difference: casual riders average 19 minutes per trip; members average 12 minutes. Casual riders take trips that are 58% longer on average, with a median of 11.3 minutes vs. 8.6 minutes for members.

Duration by Rider Type

Member average: 12 min per trip (median: 8.6 min) — short, utilitarian commutes.

Casual average: 19 min per trip (median: 11.3 min) — 58% longer, consistent with recreational use.

Casual riders extract more time per ride but pay per trip, not a flat annual fee.

Implication: a casual rider who rides frequently is paying a premium vs. what an annual membership would cover at a lower per-ride cost.

Note: Because Cyclistic did not provide pricing data in the dataset, an exact break-even calculation is not included. The structural argument holds regardless of specific fare values: a flat-fee annual membership becomes more cost-effective the more frequently a rider uses the service.

Finding 3: Casual Demand Collapses in Winter

Monthly trend analysis reveals dramatically different seasonality patterns:

Seasonal Patterns

Members maintain relatively consistent usage year-round, with moderate peaks in summer.

Casual riders nearly disappear in winter (January: ~25K rides vs. peak of ~330K in August).

Casual demand begins accelerating in April and peaks in July–August.

The gap between the two groups is largest in winter, smallest in summer.

This seasonality defines the conversion window: April through June, when casual riders are re-engaging after winter but before they have already committed to a full summer of casual usage.

Finding 4: Casual Riders Cluster at Leisure Locations

Geographic analysis of the top 100 stations per rider type reveals a clear spatial divide:

Geographic Concentration

Casual riders concentrate heavily along the lakefront — near Millennium Park, Navy Pier, and Grant Park.

These are Chicago's primary tourist and recreational corridors.

Members are distributed across residential and business districts throughout the city.

The largest casual stations by volume are tourist-adjacent, confirming the leisure profile.

This geographic intelligence informs where physical conversion touchpoints (QR codes, signage, staff) should be placed for maximum return on investment.

Recommendations

Three targeted recommendations, each directly tied to the data findings above:

1

Launch a Spring Conversion Campaign (April–June)

Casual demand is accelerating but has not yet peaked. This is the highest-ROI window to reach casual riders before they establish a full summer of pay-per-trip habits.

Offer a time-limited promotion: discounted annual membership valid for the full summer season.

Message: 'You're already riding. A membership pays for itself in a handful of trips.'

2

Target Weekend Afternoon Riders with In-App Messaging

Behavioral data shows casual riders are most active on Saturday and Sunday afternoons.

Deploy targeted in-app notifications and push messages during peak casual hours (12–6pm on weekends).

Personalize the message with the rider's actual usage data: 'You've taken 12 rides this month. An annual membership would have covered all of them at a lower per-ride cost.'

3

Concentrate Physical Conversion Touchpoints at Lakefront Stations

The top casual stations are concentrated near Millennium Park, Navy Pier, and the Museum Campus.

Place QR codes and promotional signage at dock stations in these locations.

Staff seasonal brand activations at the highest-volume casual stations during peak summer weekends.

These locations guarantee exposure to the exact target audience: high-frequency casual riders.

Conclusion

The data tells a coherent story: casual riders are leisure users, not accidental commuters. Their behavioral profile — weekends, afternoons, lakefront stations, longer trips — defines both who they are and how to reach them.

The conversion opportunity is real and time-sensitive. Casual demand spikes in summer and collapses in winter. A campaign that reaches leisure riders in spring, with a financially compelling membership offer, personalized to their actual usage, and placed at the exact stations where they ride, is not a generic marketing push. It is a precision strategy built directly from 5.7 million data points.

Interactive Dashboard: [Cyclistic Case Study on Tableau Public](#)

R Analysis Script: `cyclistic_analysis.R` (available on request)

References

Data Source

Motivate International Inc. (2026). Divvy trip data [Dataset]. Made available under the Divvy Data License Agreement. Retrieved from <https://divvy-tripdata.s3.amazonaws.com/index.html>

Note: This dataset is provided by Motivate International Inc. under a public license for non-commercial analytical use. The data has been anonymized — no personally identifiable information is included. "Cyclistic" is a fictional company name used in the context of the Google Data Analytics Certificate capstone project; the underlying data reflects real Divvy Bikes trip records from Chicago.

Methodology Framework

Google LLC. (2022). Google Data Analytics Professional Certificate. Coursera. Capstone project framework: "How Does a Bike-Share Navigate Speedy Success?" The analytical approach (Ask, Prepare, Process, Analyze, Share, Act) follows the structured methodology taught in this program.

Tools and Technologies

- R Core Team (2024). R: A language and environment for statistical computing. R Foundation for Statistical Computing. <https://www.r-project.org/>
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- Tableau Software (2026). Tableau Public [Version 2026.1.1]. Salesforce. <https://public.tableau.com/>